Semiconductor Physics and Devices

Lecture 3

Band Theory of a Solid

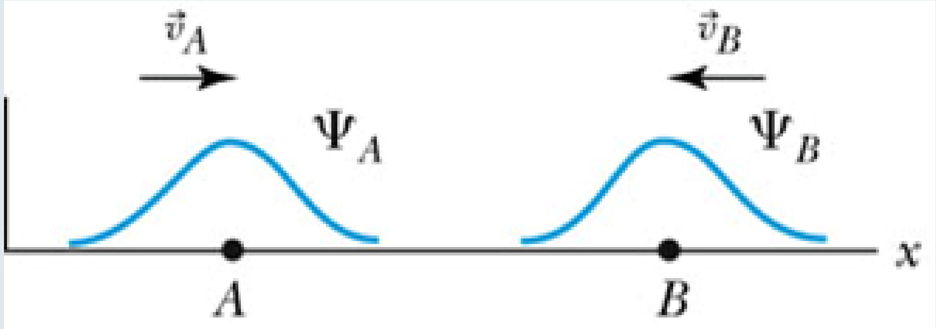
ETE-301

Band Theory of Solids

- In order to account **for *decreasing* resistivity with increasing temperature** as well as other properties of semiconductors, a new theory known as the **band theory** is introduced.
- **To understand the electrical properties of solids, the Band Theory is used.**
- The essential feature of the band theory is that the allowed **energy states for electrons are nearly continuous over certain ranges**, called **energy bands**, with forbidden energy gaps between the bands.

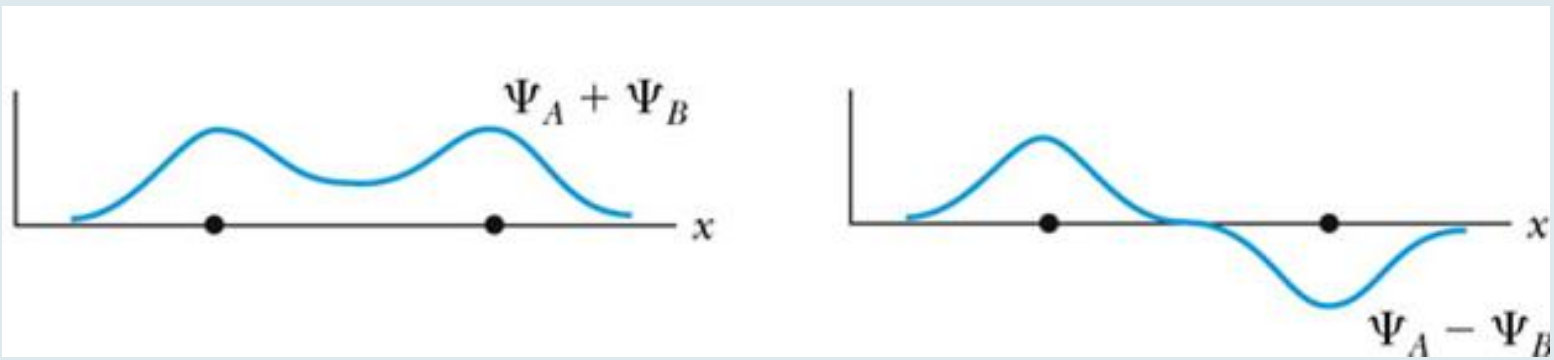
Band Theory of Solids

- Consider initially the known wave functions of two hydrogen atoms far enough apart so that they do not interact.



Band Theory of Solids

- Interaction of the wave functions occurs as the atoms get closer:



Symmetric   Antisymmetric

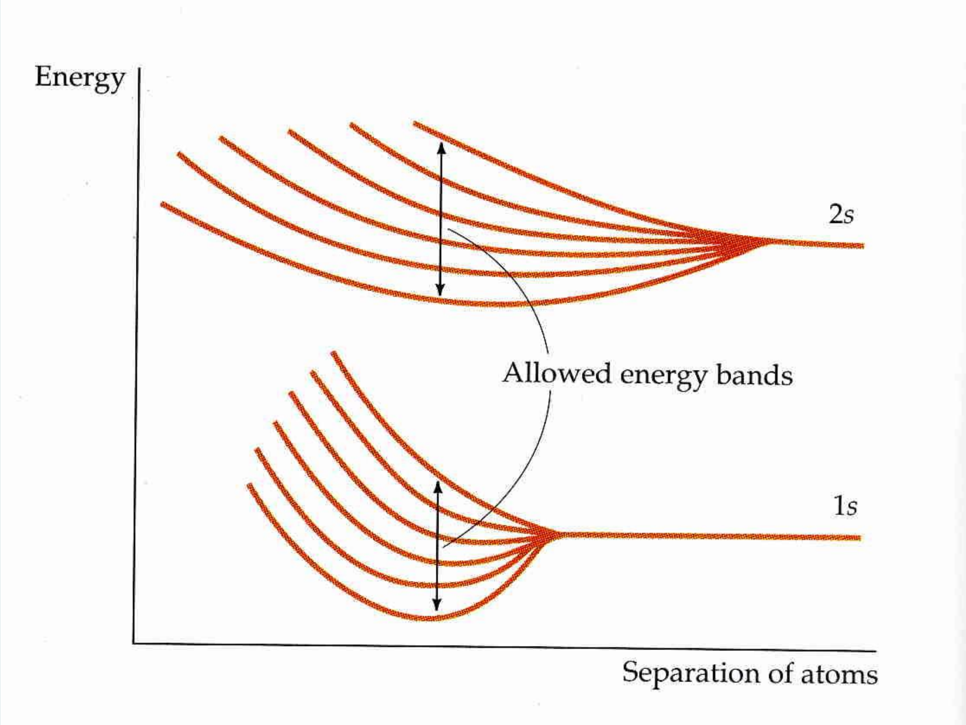
- An atom in the symmetric state has a nonzero probability of being halfway between the two atoms, while an electron in the antisymmetric state has a zero probability of being at that location.

Band Theory of Solids

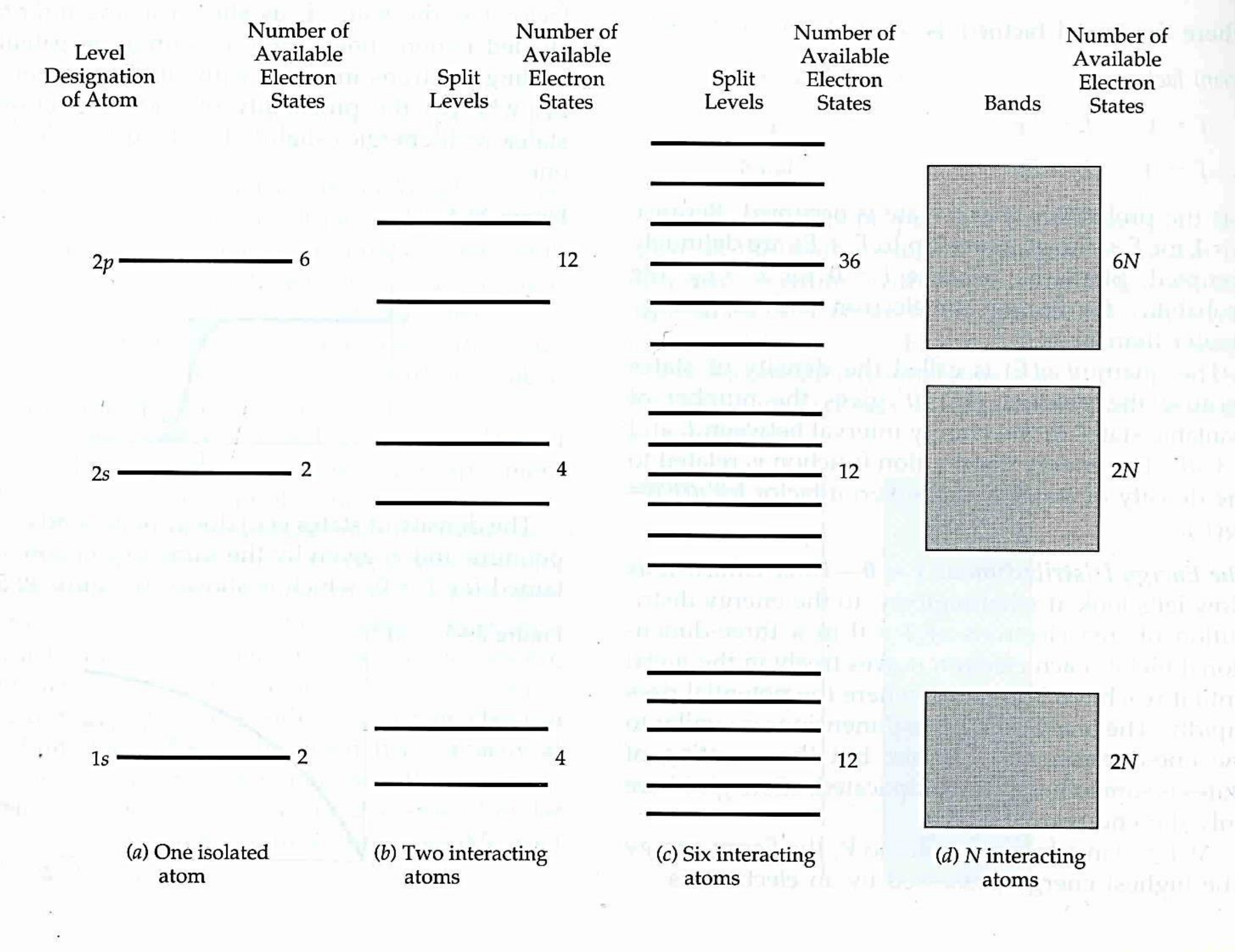
- In the symmetric case the binding energy is slightly stronger resulting in a lower energy state.
  - Thus there is a splitting of all possible energy levels (1*s*, 2*s*, and so on).

- When more atoms are added (as in a real solid), there is a further splitting of energy levels. With a large number of atoms, the levels are split into nearly continuous energy bands, with each band consisting of a number of closely spaced energy levels.

6 atoms

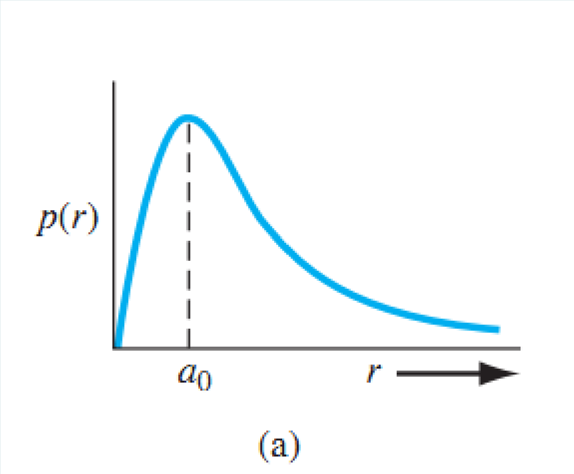


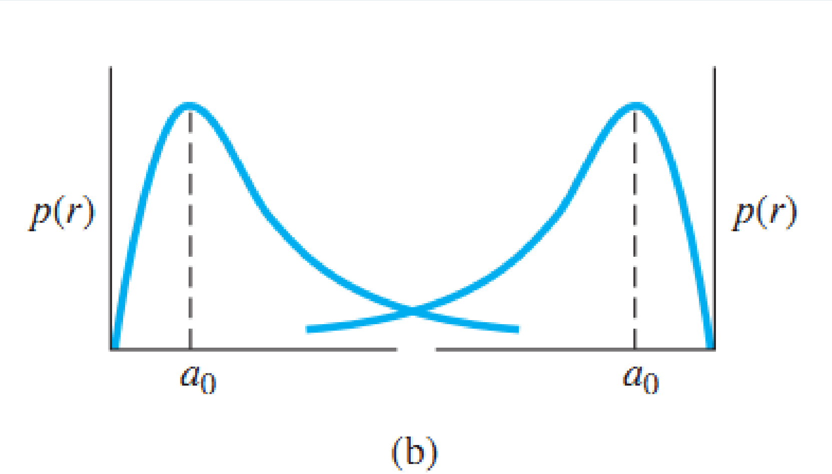
BAND STRUCTURE

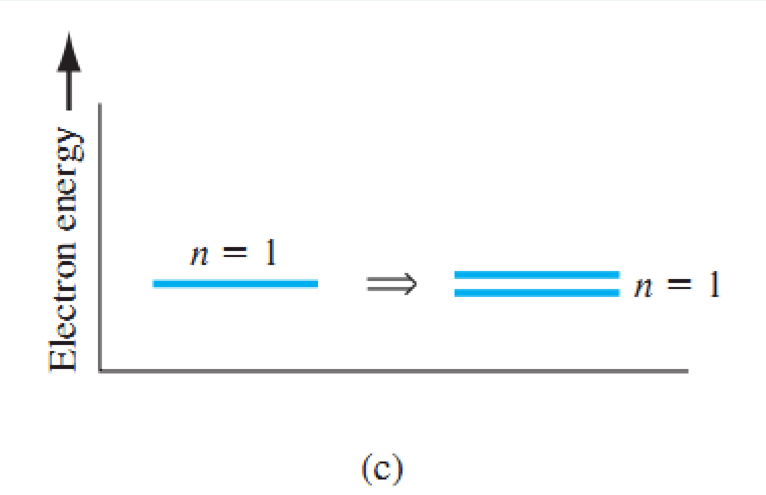


Formation of Energy Bands

the energy of the bound electron is quantized: Only discrete values of electron energy are allowed

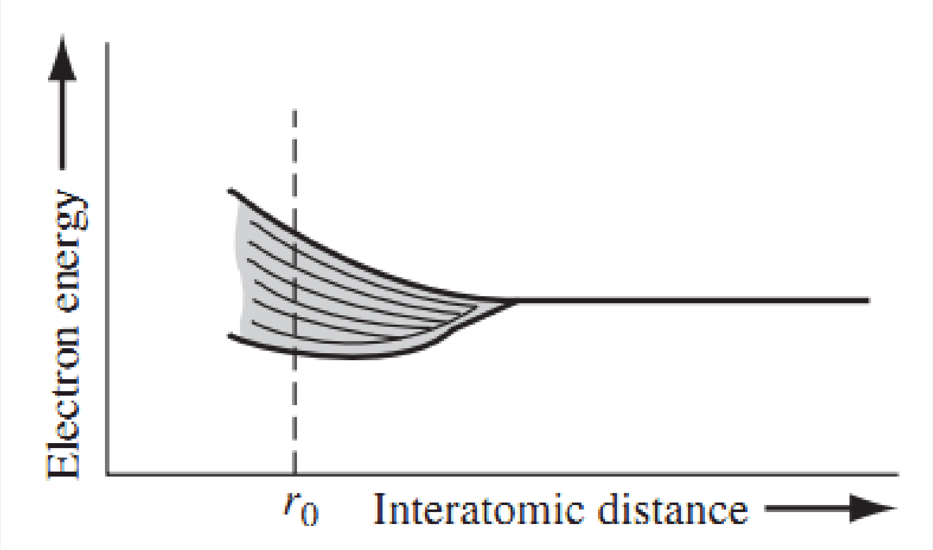






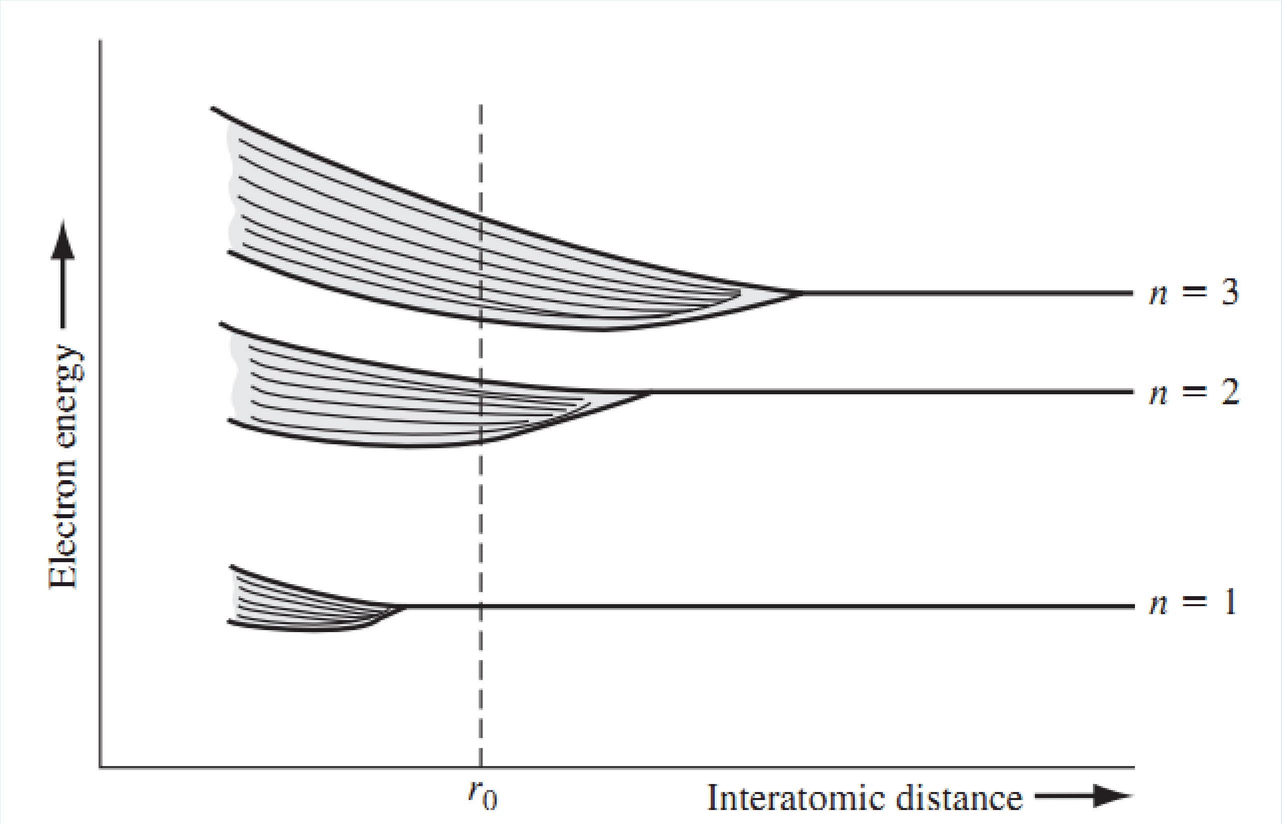
(a) Probability density function of an isolated hydrogen atom. (b) Overlapping probability density functions of two adjacent hydrogen atoms. (c) The splitting of the n=1 state.

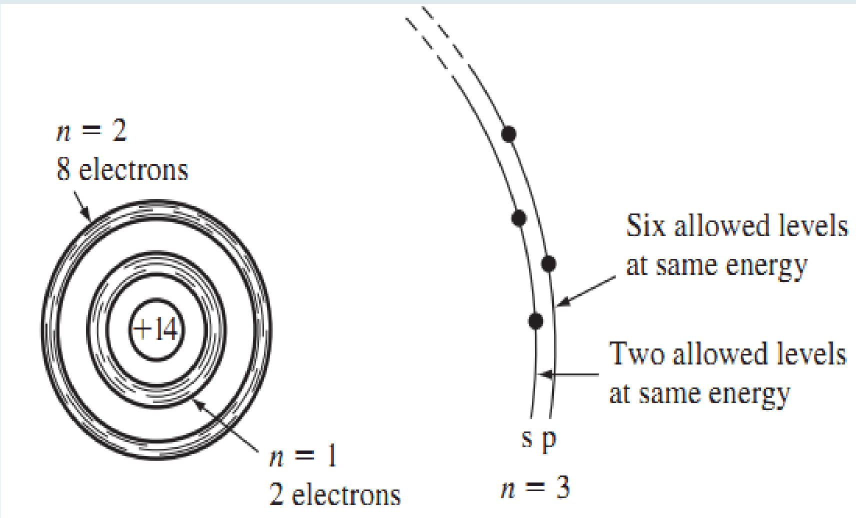
The splitting of an energy state



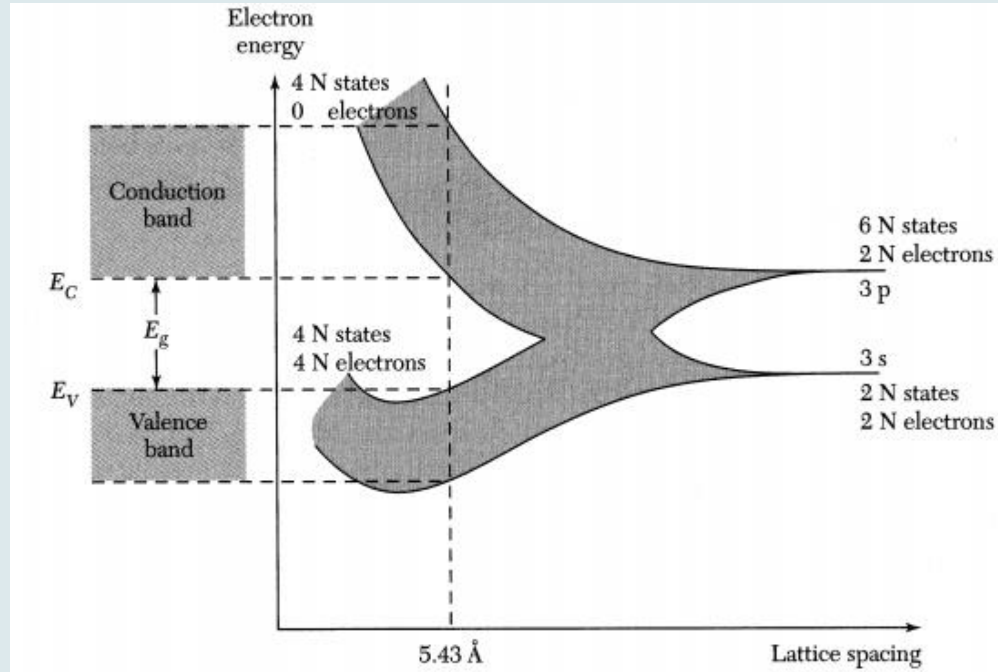
The Pauli exclusion principle states that the joining of atoms to form a system (crystal) does not alter the total number of quantum states regardless of size

the splitting of three energy states into allowed bands of energies





(a)



(a) Schematic of an isolated silicon atom. (b) The splitting of the 3s and 3p states of silicon into the allowed and forbidden energy bands.

The Energy-Momentum Diagram

The energy E of a free electron is given by

$$E = \frac{p^2}{2m_0}, \qquad \text{..................(1)}$$

where p is the momentum and *m,* is the free-electron mass. If we plot E vs. p, we obtain a parabola as shown in Fig.

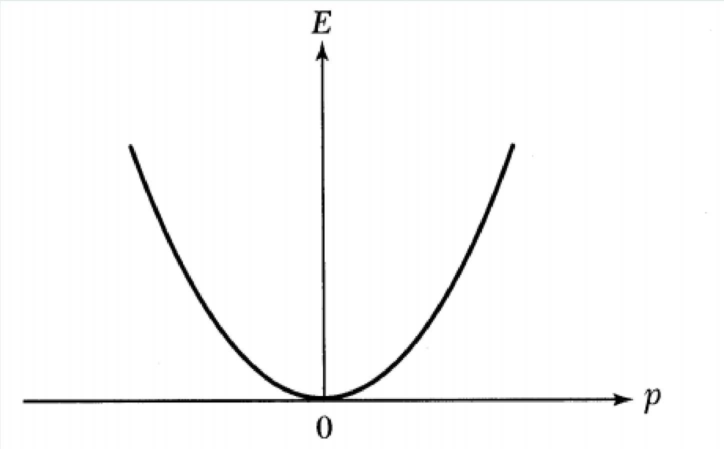


Figure: The parabolic energy (E) vs, momentum *(p)* curve for a free electron

because of the periodic potential of the nuclei, Eq. 1 can no longer be valid

Md. Ahasan Kabir, Dept. of ETE, CUET

The Energy-Momentum Diagram

replace the free-electron mass in Eq. 6 by an effective mass *m,* that is

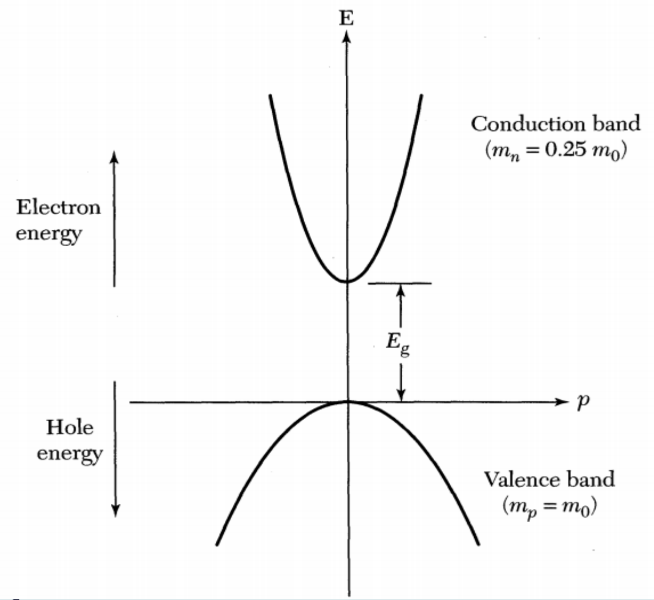
$$E = \frac{p^2}{2m_n}. \qquad \text{.....(2)}$$

The electron effective mass depends on the properties of the semiconductor. we can obtain the effective mass from the second derivative of E with respect to p:

$$m_n \equiv \left(\frac{d^2E}{dp^2}\right)^{-1}. \qquad \text{.........(3)}$$

the narrower the parabola, corresponding to a larger second derivative, the smaller the effective mass

Figure shows a simplified energy-momentum relationship of a special semiconductor with an electron effective mass of $m_n = 0.25m_0$ in the conduction band (the upper parabola) and a hole effective mass of $m_p = m_0$ in the valence band (the lower parabola).



Energy band structures of Si and GaAs